

CM12004 – Discrete Maths & Databases

Databases Coursework

<u>Set:</u> 20/11/2024 (week 8)
<u>Due:</u> 13/12/2024 (week 11), 8pm
<u>Percentage of overall unit mark:</u> 30%
<u>Submission Location:</u> Moodle.
<u>Submission Components:</u> Completed template (available on Moodle).
<u>Submission Format:</u> .pdf or .docx file.
<u>Anonymous Marking:</u> Y

1 Overview

This coursework addresses the following learning outcome for this unit:

- Evaluate, produce and implement designs using modern database technologies.

This will be assessed through your ability to critique, improve and implement a design for a scenario-based database.

2 Tasks

You must complete the following tasks based on the scenario given in section 5 and provide the requested evidence of each in the coursework template. The three primary tasks are as follows:

1. Evaluate and improve a given database design.
2. Provide the required data definition statements to construct your database design.
3. Provide the SQL queries necessary to carry out required functions.

Each of these tasks is broken down on the next page.

Task		Description	Evidence Required	Marks
1	1.1	Evaluate the logical entity-relationship diagram given in the scenario.	Complete a table describing the issues in the provided logical entity-relationship diagram.	5
	1.2	Based on your critique construct a more appropriate logical entity relationship diagram.	Provide an illustration of your updated logical entity-relationship diagram.	10
2		Implement your logical entity-relationship diagram using SQL's Data Definition Language (DDL).	SQL code.	5
3	3.1	Write a query to return the CrewID's of all crew members with the surname 'Brimble'.	SQL query.	2
	3.2	Write a query which returns the firstname and surname of all crew members who were involved in maintenance activity for the asset with ID 101.	SQL query.	2
	3.3	Write a query to retrieve a list of assets which have no maintenance activity.	SQL query.	2
	3.4	Write a query to output the ID of each asset and the number of maintenance records for each asset in ascending order.	SQL query.	2
	3.5	Write a query to output a list of crew members and the total number of maintenance records they've entered for any asset manufactured by Hitachi.	SQL query.	2
				30

3 Feedback

Formative feedback is available in the dedicated lab sessions.

You will receive **summative feedback** on your work within 3 semester weeks of the submission deadline. The feedback will discuss your performance based on the criteria for marking, including what you did well and how specific components/sections could have been improved.

4 Academic Integrity

Your work will be checked to ensure that you have not plagiarised. For more information about the plagiarism policy at the University see: <https://library.bath.ac.uk/referencing/plagiarism>

Remember that published work that you refer to in your report should be clearly referenced in your text and listed in a bibliography section given at the end of your report. For more information see, <https://library.bath.ac.uk/referencing/new-to-referencing>

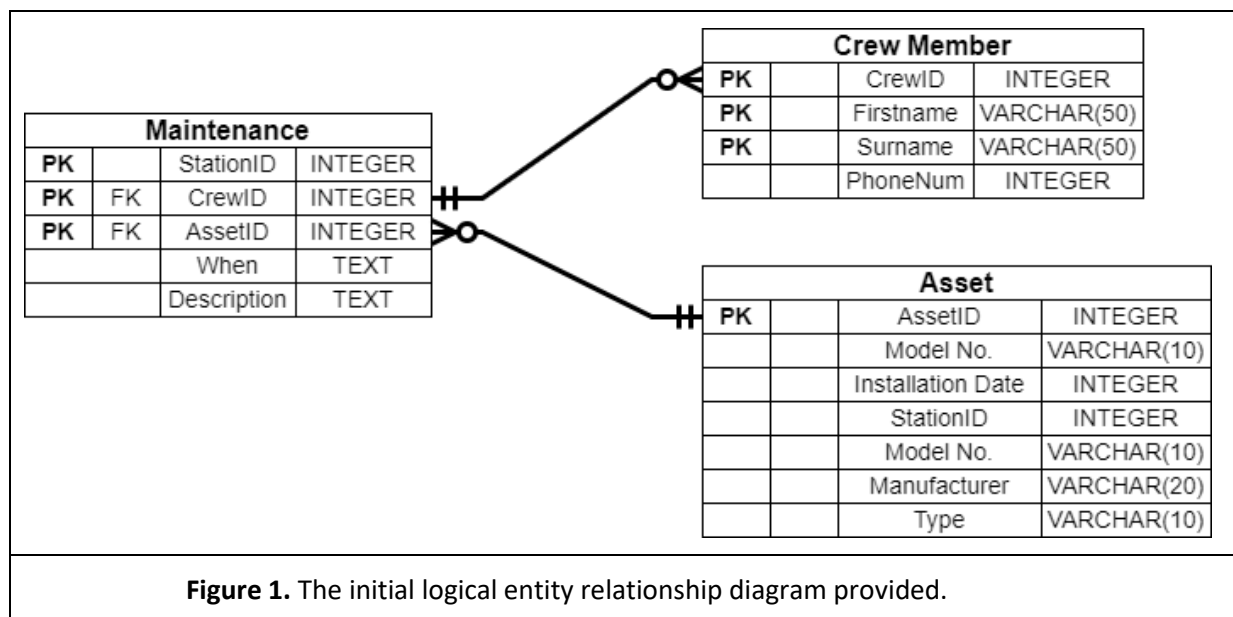
5 Scenario

A weather monitoring company WindWatchers currently run a weather station in Plymouth, this involves the collection and maintenance of a paper-based flat-file database detailing maintenance records of the station's equipment. An example of this flat file along with descriptions of its contents are given in section 6.

The company intends to expand its operation to cover many different stations in the UK; however, this will require digitalisation of their data management system. Further to this, expansion to new stations will also require storage of data relating to the current and new stations including:

- StationID: A unique identifier for each station.
- Longitude: The longitude position of the station.
- Latitude: The latitude position of the station.
- Setup Date: The day the station became operational.
- Setup Member: The ID of the crew member who led the setup of the station.

WindWatchers initially employed a local data engineer to produce a relational data model for the new system. The relational model produced consisted of an entity relationship diagram as shown in Figure 1.



Unfortunately, this entity relationship diagram was left incomplete and riddled with errors hence you have been asked to improve upon this design and provide some sample SQL statements according to the tasks presented in Section 2.

6 Example Maintenance Record File

This table shows an extract of maintenance records from the paper records.

Maintenance Records											
Date	Time	CrewID	Firstname	Surname	Phone Number	AssetID	Model No.	Manufacturer	Type	Installation Date	Description
01/02/2024	13:00	101229	Shaun	Capson	07711662883	1394	M220	Hitachi	Rainfall	03/08/2021	Replaced fuse.
01/02/2024	13:00	101938	Mellisa	Shawcross	07883999228	1019	M221	Hitachi	Rainfall	05/09/2021	Routine cleaning.
01/02/2024	14:00	101229	Shaun	Capson	07711662883	9902	KS033	Kestrel	Wind	03/08/2021	Routine cleaning.
01/02/2024	15:30	101938	Mellisa	Shawcross	07883999228	2003	PS0029	Prysmian	Pressure	01/08/2021	Removed.
01/02/2024	16:00	111002	Hugo	Morris	07655288390	1394	M220	Hitachi	Rainfall	03/08/2021	Routine cleaning.

Below are details of each attribute held in the table.

- Date: Date the maintenance record took place.
- Time: The time of the maintenance record.
- CrewID: The ID of the employee who completed the maintenance.
- Firstname: The firstname of the employee.
- Surname: The surname of the employee.
- Phone Number: The employee's phone number.
- AssetID: The ID of the piece of equipment requiring maintenance.
- Model No.: The model number of the piece of equipment.
- Manufacturer: The manufacturer of the equipment.
- Type: The type of readings which come from the piece of equipment.
- Installation Date: When the piece of equipment was installed.
- Description: Description of what maintenance took place.

7 FAQ

7.1 What should I submit?

You should submit a completed version of the template provided on Moodle, this template has space for your answers to each question.

7.2 Do I have to include code?

All code should be copied into the template. You may wish to develop your code using an SQL IDE before you copy it over to the template.

7.3 What should I use to draw my ERD?

You can use any diagram software you wish; however, the recommendation is to use 'draw.io' as this is freely available.

7.4 How many errors are there in the initial ERD?

It is difficult to put a number on the number of problems as this is based on how those problems are described. This is why you are asked to list the errors with descriptions.

7.5 Can I make assumptions?

Yes, if you are making any assumptions, please ensure they are listed in the template (space is provided for assumptions).